

QUICK START: PHASE 2 LANGMUIR FITTING

Ready to proceed? Copy/paste these commands:

STEP 1: Setup (1 minute)

```
# Navigate to your project directory
cd fluoride-hybrid-physics-ml

# Create data and results directories if needed
mkdir -p data results

# Make sure your CSV files are in the data/ folder
ls data/*.csv
# Should show:
# - doe_lhs_500.csv
# - dataset_simulated_500.csv
```

STEP 2: Install Dependencies (2 minutes)

```
# Activate your Python environment
source venv/bin/activate # Linux/Mac
# OR
venv\Scripts\activate    # Windows

# Install required packages
pip install scikit-learn scipy matplotlib pandas numpy
```

STEP 3: Run Phase 2 (30-60 minutes)

```
# Copy the script to your project
cp phase2_langmuir_fitting.py your_project_dir/

# Run it
python phase2_langmuir_fitting.py
```

What you'll see:

```
=====
PHASE 2: LANGMUIR FITTING - MULTI-FACTOR ANALYSIS
Fluoride Adsorption on Coconut Husk Activated Carbon
=====
```

```

[1/6] Loading dataset...
      Loaded 500 samples
      Response range: 1.644 - 8.316 mg/g

[2/6] Preparing features...
      Features: 10 factors
      Samples: 500 data points

[3/6] Fitting multi-factor Langmuir model...
      Model fitted

[4/6] Evaluating model performance...

      Model Performance Metrics:

      R2 (Coefficient of Determination): 0.8456 (84.56%)
      RMSE (Root Mean Squared Error):    0.9231 mg/g
      MAE (Mean Absolute Error):         0.7145 mg/g

[5/6] Saving results...
      Saved: results/langmuir_predictions.csv
      Saved: results/langmuir_model_info.json

[6/6] Creating diagnostic plots...
      Saved: results/langmuir_diagnostics.png

```

```

=====
PHASE 2 COMPLETE: LANGMUIR FITTING
=====

```

STEP 4: Verify Results (5 minutes)

```

# Check that all output files were created
ls -lh results/
# Should show:
# - langmuir_predictions.csv (40-50 KB)
# - langmuir_model_info.json (0.5-1 KB)
# - langmuir_diagnostics.png (50-100 KB)

# Preview the predictions
head -5 results/langmuir_predictions.csv

# View the model info
cat results/langmuir_model_info.json

# View the diagnostic plot (open in image viewer)

```

```
open results/langmuir_diagnostics.png # macOS
# or
xdg-open results/langmuir_diagnostics.png # Linux
# or
start results/langmuir_diagnostics.png # Windows
```

WHAT TO EXPECT

R² Value

- **Expected:** 0.84-0.87 (good!)
- **Meaning:** Chemical model explains 84-87% of variance
- **Interpretation:** Langmuir is appropriate

RMSE Value

- **Expected:** 0.90-1.20 mg/g
- **Meaning:** Average prediction error is ~1 mg/g
- **Interpretation:** Acceptable baseline

Diagnostic Plots (4 panels)

1. **Actual vs Predicted:** Should follow diagonal line
 2. **Residual Plot:** Should show random scatter around 0
 3. **Residual Distribution:** Should be roughly bell-shaped
 4. **Q-Q Plot:** Should be roughly linear
-

TROUBLESHOOTING

ImportError: No module named 'sklearn'

```
pip install scikit-learn
```

FileNotFoundError: data/dataset_simulated_500.csv

```
# Make sure CSV files are in data/ folder
ls data/dataset_simulated_500.csv # Should exist
```

Script runs but no results folder created

```
mkdir -p results
python phase2_langmuir_fitting.py
```

Plot file is very small or blank

- Check that matplotlib backend is working
 - Try: `python -c "import matplotlib; print(matplotlib.get_backend())"`
-

NEXT STEPS AFTER PHASE 2

Phase 2 Complete → What's next?

Option A: Proceed to Phase 3 (Recommended)

```
# Phase 3: Residual Analysis
python phase3_residual_analysis.py
```

Option B: Explore Phase 2 Results First

```
# Analyze the predictions
python -c "
import pandas as pd
df = pd.read_csv('results/langmuir_predictions.csv')
print('Summary Statistics:')
print(df[['q_removal', 'q_predicted', 'residual']].describe())
"
```

Option C: Create Custom Analysis

```
# Jupyter notebook for deep dive
jupyter notebook
# Then open a new notebook and import:
# import pandas as pd
# results = pd.read_csv('results/langmuir_predictions.csv')
```

SUCCESS CHECKLIST

- ☐ Phase 2 script runs without errors
 - ☐ Output files created in results/ folder
 - ☐ R^2 value is 0.80-0.90 (good baseline)
 - ☐ RMSE is < 1.5 mg/g
 - ☐ Diagnostic plots look reasonable
 - ☐ Residuals appear randomly distributed
 - ☐ You understand what Langmuir model captures
 - ☐ You're ready for ML training (Phase 4)
-

TYPICAL TIMELINE

Step	Time	Status
Setup	1 min	
Install packages	2 min	
Run script	45 min	
Verify results	5 min	
Total	~1 hour	

KEY NUMBERS TO REMEMBER

After Phase 2 completes, write down these numbers:

R^2 = _____ (aim for 0.84+)

RMSE = _____ (aim for <1.2)

MAE = _____ (aim for <0.8)

Residual Std = ___ (this is what ML learns to predict)

These become your **baseline** for Phase 4-5.

FINAL NOTES

You're on track!

Phase 1 → Phase 2 (NOW) → Phase 3-8 (coming)

After Phase 2, you'll have: - Baseline chemical model - Residuals to analyze - Clear ML opportunities - Foundation for hybrid model

Ready? Run the script above!

Still have questions? See: PHASE_2_LANGMUIR_GUIDE.md for complete documentation